

## G76 Threading Cycle, Multiple Pass (Group 00)

\***A** - Tool nose angle (value: 0 to 120 degrees) Do not use a decimal point

**D** - First pass cutting depth

**F(E)** - Feed rate, the lead of the thread

\***I** - Thread taper amount, radius measure

**K** - Thread height, defines thread depth, radius measure

\***P** - Single Edge Cutting (load constant)

\***Q** - Thread Start Angle (Do not use a decimal point)

\***U** - X-axis incremental distance, start to maximum thread Depth Diameter

\***W** - Z-axis incremental distance, start to maximum thread length

\***X** - X-axis absolute location, maximum thread Depth Diameter

\***Z** - Z-axis absolute location, maximum thread length

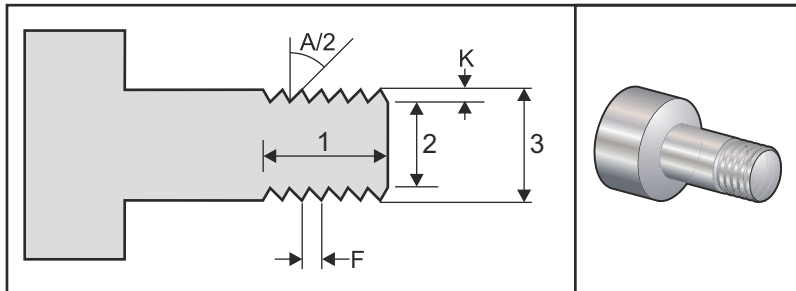
\* indicates optional



### NOTE:

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

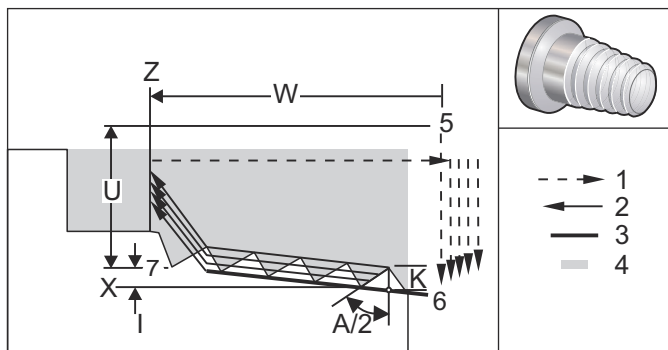
**F7.32:** G76 Threading Cycle, Multiple Pass: [1] Z depth, [2] Minor diameter, [3] Major diameter.



Setting 95/Setting 96 determine chamfer size/angle; M23/M24 turn chamfering **ON/OFF**.

**F7.33:**

G76 Threading Cycle, Multiple Pass Tapered: [1] Rapid, [2] Feed, [3] Programmed path, [4] Cut allowance, [5] Start position, [6] Finished diameter, [7] Target, [A] Angle.



The G76 canned cycle can be used for threading both straight or tapered (pipe) threads.

The height of the thread is defined as the distance from the crest of the thread to the root of the thread. The calculated depth of thread ( $K$ ) is the value of  $K$  less the finish allowance (Setting 86, Thread Finish Allowance).

The thread taper amount is specified in  $I$ . Thread taper is measured from the target position  $X$ ,  $Z$  at point [7] to position [6]. The  $I$  value is the difference in radial distance from the start to the end of the thread, not an angle.

**NOTE:**

*A conventional O.D. taper thread will have a negative  $I$  value.*

The depth of the first cut through the thread is specified in  $D$ . The depth of the last cut through the thread can be controlled with Setting 86.

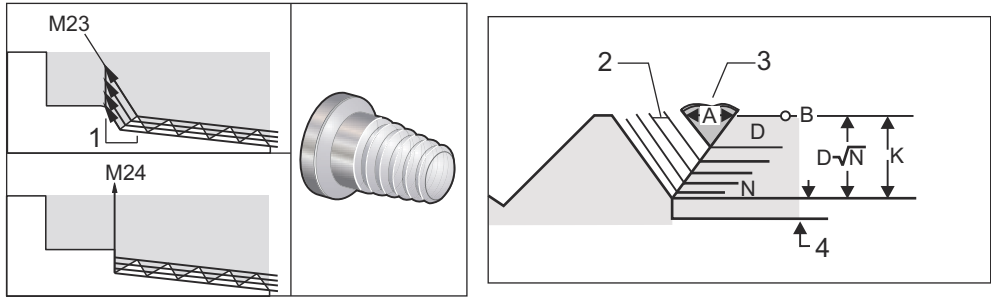
The tool nose angle for the thread is specified in  $A$ . The value can range from 0 to 120 degrees. If  $A$  is not used, 0 degrees is assumed. To reduce chatter while threading use  $A59$  when cutting a 60 degree included thread.

The  $F$  code specifies the feed rate for threading. It is always good programming practice to specify  $G99$  (feed per revolution) prior to a threading canned cycle. The  $F$  code also indicates the thread pitch or lead.

At the end of the thread an optional chamfer is performed. The size and angle of the chamfer is controlled with Setting 95 (Thread Chamfer Size) and Setting 96 (Thread Chamfer Angle). The chamfer size is designated in number of threads, so that if 1.000 is recorded in Setting 95 and the feed rate is .05, then the chamfer will be .05. A chamfer can improve the appearance and functionality of threads that must be machined up to a shoulder. If relief is provided for at the end of the thread then the chamfer can be eliminated by specifying 0.000 for the chamfer size in Setting 95, or using  $M24$ . The default value for Setting 95 is 1.000 and the default angle for the thread (Setting 96) is 45 degrees.

**F7.34:**

G76 Using an A Value: [1] Setting 95 and 96 (see Note), [2] Setting 99 (Thread Minimum Cut), [3] Cutting Tip, [4] Setting 86 - Finish Allowance.

**NOTE:**

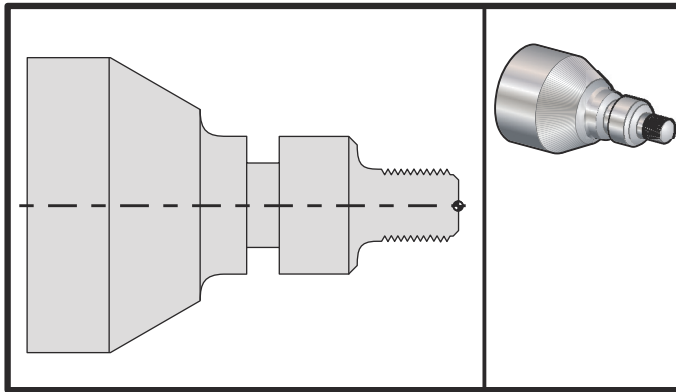
*Setting 95 and 96 will affect the final chamfer size and angle.*

Four options for G76 Multiple Thread Cutting are available:

1. P1: Single edge cutting, cutting amount constant
2. P2: Double edge cutting, cutting amount constant
3. P3: Single edge cutting, cutting depth constant
4. P4: Double edge cutting, cutting depth constant

P1 and P3 both allow for single edge threading, but the difference is that with P3 a constant depth cut is done with every pass. Similarly, P2 and P4 options allow for double edge cutting with P4 giving constant depth cut with every pass. Based on industry experience, double edge cutting option P2 may give superior threading results.

D specifies the depth of the first cut. Each successive cut is determined by the equation  $D \cdot \sqrt{N}$  where N is the Nth pass along the thread. The leading edge of the cutter does all of the cutting. To calculate the X position of each pass you have to take the sum of all the previous passes, measured from the start point the X value of each pass

**F7.35:****G76 Thread Cutting Cycle, Multiple Pass**

```

%
o60761 (G76 THREAD CUTTING MULTIPLE PASSES) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD thread tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X1.2 Z0.3 (Rapid to 1st position) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G76 X0.913 Z-0.85 K0.042 D0.0115 F0.0714 (Begin G76) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

**G80 Canned Cycle Cancel (Group 09)**

G80 cancels all active canned cycles.

**NOTE:**

*G00 or G01 also cancel canned cycles.*

## G81 Drill Canned Cycle (Group 09)

\***C** - C-Axis absolute motion command (optional)

**F** - Feed Rate

\***L** - Number of repeats

**R** - Position of the R plane

\***X** - X-axis motion command

\***Y** - Y-axis absolute motion command

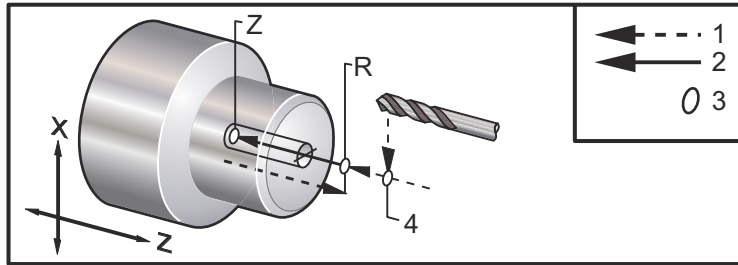
**Z** - Position of bottom of hole

\* indicated optional

Also see G241 for radial drilling and G195/G196 for radial tapping with live tooling.

### F7.36:

G81 Drill Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting plane, [R] R plane, [Z] Position at the bottom of the hole.



## G82 Spot Drill Canned Cycle (Group 09)

\***C** - C-Axis absolute motion command (optional)

**F** - Feed Rate in inches (mm) per minute

\***L** - Number of repeats

**P** - The dwell time at the bottom of the hole

**R** - Position of the R plane

\***X** - X-axis motion command

\***Y** - Y-axis motion command

**Z** - Position of bottom of hole

\* indicates optional

This G code is modal in that it activates the canned cycle until it is canceled or another canned cycle is selected. Once activated, every motion of X will cause this canned cycle to be executed.

Also, see G242 for radial live tool spot drilling.

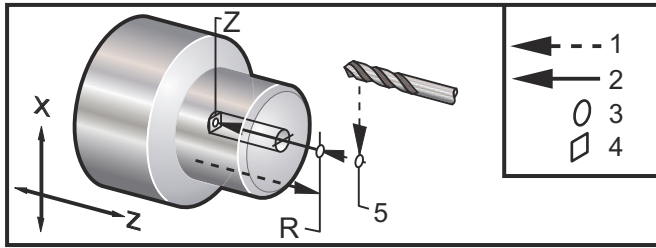


### NOTE:

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

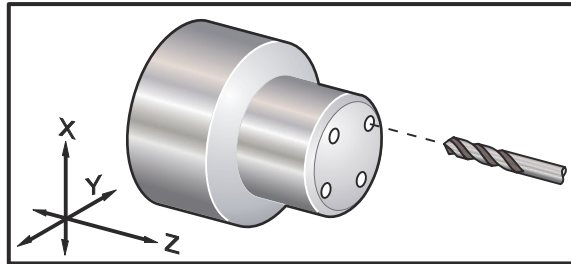
**F7.37:**

G82 Spot Drill Canned Cycle:[1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Dwell, [5] Starting plane, [R] R plane, [Z] Position of the bottom of the hole.



**F7.38:**

G82 Y-Axis Drill



```
%  
o60821 (G82 LIVE SPOT DRILL CYCLE) ;  
(G54 X0 Y0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is a spot drill) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
G98 (Feed per min) ;  
M154 (Engage C Axis) ;  
G00 G54 X1.5 C0. Z1. (Rapid to 1st position) ;  
P1500 M133 (Live tool CW at 1500 RPM) ;  
M08 (coolant on) ;  
(BEGIN CUTTING CYCLE) ;  
G82 C45. Z-0.25 F10. P80 (Begin G82) ;  
C135. (2nd position) ;  
C225. (3rd position) ;  
C315. (4th position) ;  
(BEGIN COMPLETION BLOCKS) ;  
M155 (C axis disengage) ;  
M135 (Live tool off) ;  
G00 G53 X0 M09 (X home, coolant off) ;  
G53 Z0 (Z home) ;  
M30 (End program) ;
```

To calculate how long you should dwell at the bottom of your spot drill cycle, use the following formula:

$$P = \text{Dwell Revolutions} \times 60000/\text{RPM}$$

If you want the tool to dwell for two full revolutions at its full Z depth in the program above (running at 1500 RPM), you would calculate:

$$2 \times 60000 / 1500 = 80$$

Enter P80 (80 milliseconds or P.08 (.08 seconds) on the G82 line, to dwell for 2 revolutions at 1500 RPM.

## G83 Normal Peck Drilling Canned Cycle (Group 09)

\*C - C-Axis absolute motion command (optional)

F - Feed Rate in inches (mm) per minute

\*I - Size of first cutting depth

\*J - Amount to reduce cutting depth each pass

\*K - Minimum depth of cut

\*L - Number of repeats

\*P - The dwell time at the bottom of the hole

\*Q - The cut-in value, always incremental

\*R - Position of the R plane

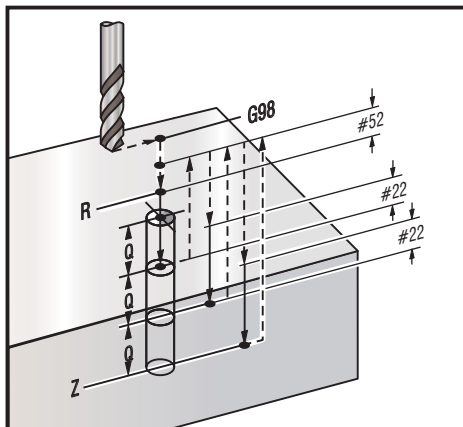
\*X - X-axis motion command

\*Y - Y-axis motion command

Z - Position of bottom of hole

\* indicates optional

**F7.39:** G83 Peck Drilling Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Dwell, [#22] Setting 22, [#52] Setting 52.



**NOTE:**

*If  $I$ ,  $J$ , and  $K$  are specified, a different operating mode is selected. The first pass will cut in the value of  $I$ , each succeeding cut will be reduced by amount  $J$ , and the minimum cutting depth is  $K$ . Do not use a  $Q$  value when programming with  $I$ ,  $J$ , and  $K$ .*

Setting 52 changes the way G83 works when it returns to the  $R$  plane. Usually the  $R$  plane is set well outside the cut to insure that the chip clearing motion allows the chips to clear the hole. However, this is wasted motion when first drilling through this empty space. If Setting 52 is set to the distance required to clear chips, the  $R$  plane can be put much closer to the part being drilled. When the clear move to  $R$  occurs, the  $Z$  will be moved past  $R$  by this value in Setting 52. Setting 22 is the amount to feed in  $Z$  to get back to the same point at which the retraction occurred.

```
%  
o60831 (G83 NORMAL PECK DRILLING) ;  
(G54 X0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is a drill) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
G50 S1000 (Limit spindle to 1000 RPM) ;  
G97 S500 M03 (CSS off, spindle on CW) ;  
G00 G54 X0 Z0.25 (Rapid to 1st position) ;  
M08 (Coolant on) ;  
(BEGIN CUTTING BLOCKS) ;  
G83 Z-1.5 F0.005 Q0.25 R0.1 (Begin G83)  
(BEGIN COMPLETION BLOCKS)  
G00 G53 X0 M09 (X home, coolant off) ;  
G53 Z0 M05 (Z home, spindle off) ;  
M30 ;  
%
```

```
%  
(LIVE PECK DRILL - AXIAL) ;  
T1111 ;  
G98 ;  
M154 (Engage C-Axis) ;  
G00 G54 X6. C0. Y0. Z1. ;  
G00 X1.5 Z0.25 ;  
G97 P1500 M133 ;  
M08 ;  
G83 G98 C45. Z-0.8627 F10. Q0.125 ;
```



```

C135. ;
C225. ;
C315. ;
G00 G80 Z0.25 ;
M155 ;
M135 ;
M09 ;
G28 H0. (Unwind C-Axis) ;
G00 G54 X6. Y0. Z1. ;
G18 ;
G99 ;
M01 ;
M30 ;
%
```

## G84 Tapping Canned Cycle (Group 09)

**F** - Feed Rate

\* **R** - Position of the R plane

**S** - RPM, called prior to G84

\* **X** - X-axis motion command

**Z** - Position of bottom of hole

\* indicates optional

Programming Notes:

- It is not necessary to start the spindle CW before this canned cycle. The control does this automatically.
- When G84 tapping on a lathe, it is simplest to use G99 Feed Per Revolution.
- The Lead is the distance traveled along a screw's axis, with each full revolution.
- The feedrate, when using G99, is equal to the Lead of the tap.
- An S value must be called prior to the G84. The S value determines the RPM of the tapping cycle.
- In Metric Mode ( G99, with Setting 9 = **MM**), the feedrate is the metric equivalent of the lead, in **MM**.
- In Inch Mode ( G99, with Setting 9 = **INCH**), the feedrate is the Inch equivalent of the lead, in inches.
- The lead (and G99 feedrate) of an M10 x 1.0mm tap is 1.0mm, or .03937" (1.0/25.4=.03937).

Examples:

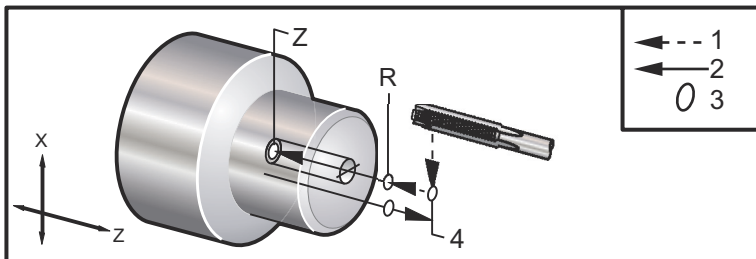
1. The lead of a 5/16-18 tap is 1.411 mm ( $1/18 \times 25.4 = 1.411$ ), or .0556" ( $1/18 = .0556$ )
2. This canned cycle can be used on the secondary spindle of a Dual Spindle DS lathe, when prefaced by a G14.  
Refer to the G14 Secondary Spindle Swap on page **275** for more information.

3. For Axial Live-Tool tapping, use a G95 or G186 command.
4. For Radial Live-Tool tapping, use a G195 or G196 command.
5. For Reverse Tapping (left-hand thread) on the Main or Secondary Spindle, refer to page **323**.

More programming examples, in both Inch and Metric, are shown below:

**F7.40:**

G84 Tapping Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting plane, [R] R plane, [Z] Position at the bottom of the hole.



```
%
o60841 (IMPERIAL TAP, SETTING 9 = MM) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part)
(T1 is a 1/4-20 Tap) ;
G21 (ALARM if setting 9 is not MM) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G40 G80 G99 (Safe startup) ;
G00 G54 X0 Z12.7 (Rapid to 1st position) ;
M08 (Coolant on) ;
S800 (RPM OF TAP CYCLE) ;
(BEGIN CUTTING BLOCK) ;
G84 Z-12.7 R12.7 F1.27 (1/20*25.4 = 1.27) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

```
%
o60842 (METRIC TAP, SETTING 9 = MM) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part)
(T1 is an M8 x 1.25 Tap) ;
G21 (ALARM if setting 9 is not MM) ;
(BEGIN PREPARATION BLOCKS) ;
```

```

T101 (Select tool and offset 1) ;
G00 G18 G40 G80 G99 (Safe startup) ;
G00 G54 X0 Z12.7 (Rapid to 1st position) ;
M08 (Coolant on) ;
S800 (RPM OF TAP CYCLE) ;
(BEGIN CUTTING BLOCK) ;
G84 Z-12.7 R12.7 F1.25 (Lead = 1.25) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

```

%
o60843 (IMPERIAL TAP, SETTING 9 = IN) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part)
(T1 is a 1/4-20 Tap) ;
G20 (ALARM if setting 9 is not INCH) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G00 G54 X0 Z0.5 (Rapid to 1st position) ;
M08 (Coolant on) ;
S800 (RPM OF TAP CYCLE) ;
(BEGIN CUTTING BLOCK) ;
G84 Z-0.5 R0.5 F0.05 (Begin G84) ;
(1/20 = .05) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

```

%
o60844 (METRIC TAP, SETTING 9 = IN) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part)
(T1 is an M8 x 1.25 Tap) ;
G20 (ALARM if setting 9 is not INCH) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G00 G54 X0 Z0.5 (Rapid to 1st position) ;
M08 (Coolant on) ;
```

```

S800 (RPM OF TAP CYCLE) ;
(BEGIN CUTTING BLOCK) ;
G84 Z-0.5 R0.5 F0.0492 (1.25/25.4 = .0492) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

## G85 Boring Canned Cycle (Group 09)

 **NOTE:** *This cycle feeds in and feeds out.*

**F** - Feed Rate

**\*L** - Number of repeats

**\*R** - Position of the R plane

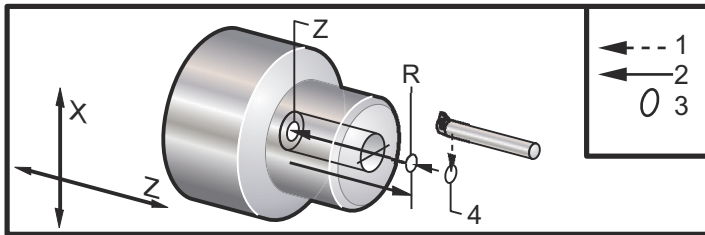
**\*X** - X-axis motion command

**\*Y** - Y-axis motion command

**Z** - Position of bottom of hole

\* indicates optional

**F7.41:** G85 Boring Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting plane, [R] R plane, [Z] Position of the bottom of the hole.



## G86 Bore and Stop Canned Cycle (Group 09)



**NOTE:**

*The spindle stops and it rapids out of the hole.*

**F** - Feed Rate

**\*L** - Number of repeats

**\*R** - Position of the R plane

**\*X** - X-axis motion command

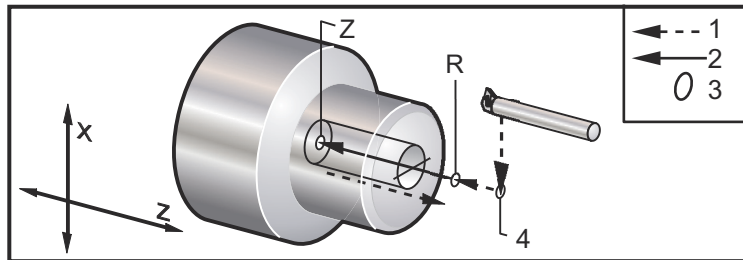
**\*Y** - Y-axis motion command

**Z** - Position of bottom of hole

\* indicates optional

This G code stops the spindle once the tool reaches the bottom of the hole. The tool retracts once the spindle has stopped.

**F7.42:** G86 Bore and Stop Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting plane, [R] R plane, [Z] Position of the bottom of the hole.



## G89 Bore and Dwell Canned Cycle (Group 09)



**NOTE:**

*This cycle feeds in and feeds out.*

**F** - Feed Rate

**\*L** - Number of repeats

**\*P** - The dwell time at the bottom of the hole

**\*R** - Position of the R plane

**\*X** - X-axis motion command

**\*Y** - Y-axis motion command

**Z** - Position of bottom of hole

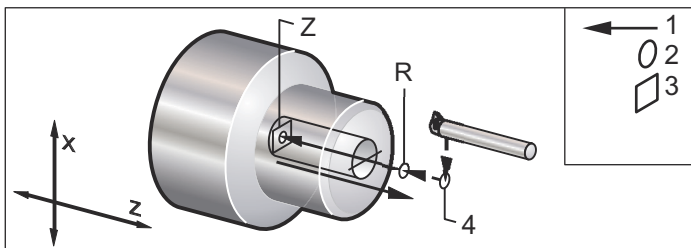
\* indicates optional

**NOTE:**

The *P* values are modal. This means if you are in the middle of a canned cycle and a *G04 Pnn* or an *M97 Pnn* is used the *P* value will be used for the dwell / subprogram as well as the canned cycle.

**F7.43:**

G89 Bore and Dwell Canned Cycle: [1] Feed, [2] Start or end of stroke, [3] Dwell, [4] Starting plane, [R] R plane, [Z] Position of the bottom of the hole.

**G90 O.D./I.D. Turning Cycle (Group 01)**

**F(E)** - Feed rate

\***I** - Optional distance and direction of X Axis taper, radius

\***U** - X-axis incremental distance to target, diameter

\***W** - Z-axis incremental distance to target

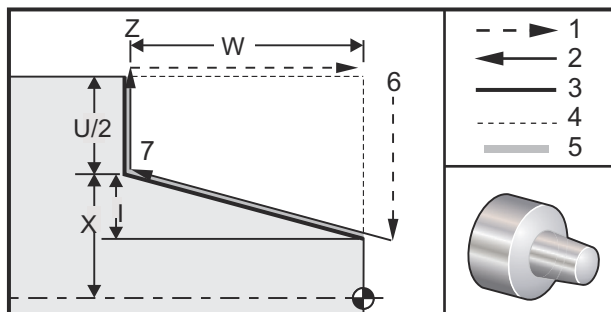
**X** - X-axis absolute location of target

**Z** - Z-axis absolute location of target

\*indicates optional

**F7.44:**

G90 O.D./I.D. Turning Cycle: [1] Rapid, [2] Feed, [3] Programmed path, [4] Cut allowance, [5] Finish allowance, [6] Start position, [7] Target.

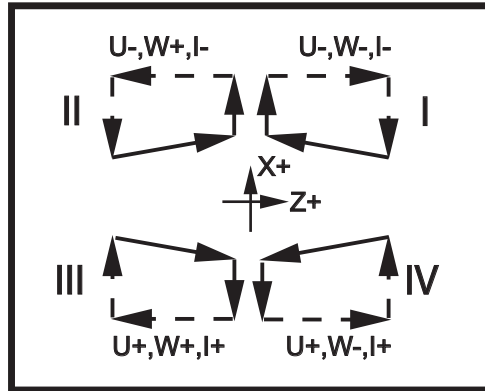


G90 is used for simple turning, however, multiple passes are possible by specifying the *X* locations of additional passes.

Straight turning cuts are made by specifying *X*, *Z* and *F*. By adding an *I* value, a taper cut is made. The amount of taper is referenced from the target. That is, *I* is added to the value of *X* at the target.

Any of the four ZX quadrants can be programmed using U, W, X, and Z; the taper is positive or negative. The following figure gives a few examples of the values required for machining in each of the four quadrants.

**F7.45:** G90-G92 Address Relationships



## G92 Threading Cycle (Group 01)

**F(E)** - Feed rate, the lead of the thread

**\*I** - Optional distance and direction of X Axis taper, radius

**\*Q** - Start Thread Angle

**\*U** - X-axis incremental distance to target, diameter

**\*W** - Z-axis incremental distance to target

**X** - X-axis absolute location of target

**Z** - Z-axis absolute location of target

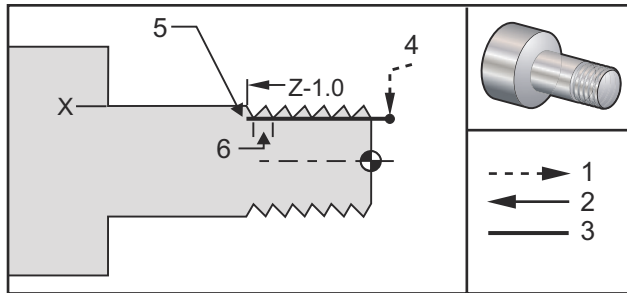
\* indicates optional

### Programming Notes:

- Setting 95/Setting 96 determine chamfer size/angle. M23/M24 turn chamfering on/off.
- G92 is used for simple threading, however, multiple passes for threading are possible by specifying the X locations of additional passes. Straight threads are made by specifying X, Z, and F. By adding an I value, a pipe or taper thread is cut. The amount of taper is referenced from the target. That is, I is added to the value of X at the target. At the end of the thread, an automatic chamfer is cut before reaching the target; default for this chamfer is one thread at 45 degrees. These values can be changed with Setting 95 and Setting 96.
- During incremental programming, the sign of the number following the U and W variables depends on the direction of the tool path. For example, if the direction of a path along the X-axis is negative, the value of U is negative.

**F7.46:**

G92 Threading Cycle: [1] Rapid, [2] Feed, [3] Programmed path, [4] Start position, [5] Minor diameter, [6] 1/Threads per inch = Feed per revolution (Inch formula; F = lead of thread) .



```

%
O60921 (G92 THREADING CYCLE) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD thread tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X0 Z0.25 (Rapid to 1st position) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
X1.2 Z.2 (Rapid to clear position) ;
G92 X.980 Z-1.0 F0.0833 (Begin Thread Cycle) ;
X.965 (2nd pass) ;
X.955 (3rd pass) ;
X.945 (4th pass) ;
X.935 (5th pass) ;
X.925 (6th pass) ;
X.917 (7th pass) ;
X.910 (8th pass) ;
X.905 (9th pass) ;
X.901 (10th pass) ;
X.899 (11th pass) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```



## G94 End Facing Cycle (Group 01)

**F(E)** - Feed rate

\***K** - Optional distance and direction of Z Axis coning

\***U** - X-axis incremental distance to target, diameter

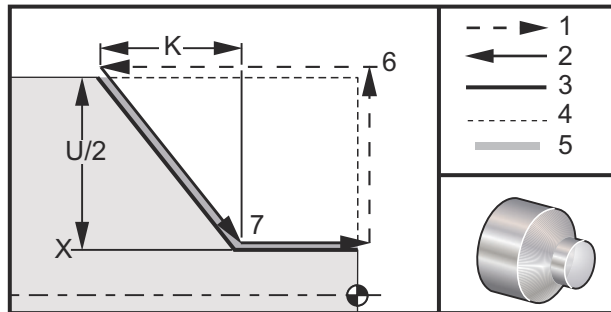
\***W** - Z-axis incremental distance to target

**X** - X-axis absolute location of target

**Z** - Z-axis absolute location of target

\*indicates optional

**F7.47:** G94 End Facing Cycle: [1] Rapid, [2] Feed, [3] Programmed path, [4] Cut allowance, [5] Finish allowance, [6] Start position, [7] Target.

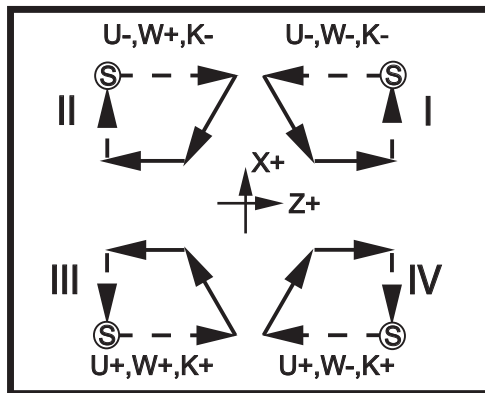


Straight end facing cuts can be made by specifying  $X$ ,  $Z$  and  $F$ . By adding  $K$  a cone-shaped face is cut. The amount of coning is referenced from the target. That is  $K$  is added to the value of  $X$  at the target.

Any of the four ZX quadrants is programmed by varying  $U$ ,  $W$ ,  $X$ , and  $Z$ . The coning is positive or negative. The following figure gives a few examples of the values required for machining in each of the four quadrants.

During incremental programming, the sign of the number following the  $U$  and  $W$  variables depends on the direction of the tool path. If the direction of a path along the X-axis is negative, the value of  $U$  is negative.

**F7.48:** G94 Address Relationships: [S] Start position.



## G95 Live Tooling Rigid Tap (Face) (Group 09)

**\*C** - C-Axis absolute motion command (optional)

**F** - Feed Rate

**R** - Position of the R plane

**S** - RPM, called prior to G95

**W** - Z-axis incremental distance

**X** - Optional Part Diameter X-axis motion command

**\*Y** - Y-axis motion command

**Z** - Position of bottom of hole

\* indicates optional

G95 Live Tooling Rigid Tapping is an axial tapping cycle similar to G84 Rigid Tapping in that it uses the F, R, X and Z addresses, however, it has the following differences:

- The control must be in G99 Feed per Revolution mode in order for tapping to work properly.
- An S (spindle speed) command must have been issued prior to the G95.
- The X Axis must be positioned between machine zero and the center of the main spindle, do not position beyond spindle center.

```
%  
o60951 (G95 LIVE TOOLING RIGID TAP) ;  
(G54 X0 Y0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is a 1/4-20 tap) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
M154 (Engage C Axis) ;  
G00 G54 X1.5 C0. Z0.5 (Rapid to 1st position) ;  
M08 (Coolant on) ;
```

```

(BEGIN CUTTING CYCLE) ;
S500 (Select tap RPM) ;
G95 C45. Z-0.5 R0.5 F0.05 (Tap to Z-0.5) ;
C135. (next position) ;
C225. (next position) ;
C315. (last position) ;
(BEGIN COMPLETION BLOCKS) ;
M155 (Disengage C Axis) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 (Z home) ;
M30 (End program) ;
%
```

## G96 Constant Surface Speed ON (Group 13)

G96 commands the control to maintain a constant cutting speed at the tip of the tool. The spindle RPM is based on the diameter of the part where the cut is taking place, and the commanded **S** value ( $\text{RPM} = 3.82 \times \text{SFM} / \text{DIA}$ ). This means the spindle speed increases as the tool gets closer to **X0**. When Setting 9 is set to **INCH**, the **S** value specifies Surface Feet Per Minute. When Setting 9 is set to **MM**, the **S** value specifies Surface Meters Per Minute.



### **WARNING:**

*It is safest to specify a maximum spindle speed for the Constant Surface Speed feature. Use G50 to set a maximum spindle RPM. Not setting a limit allows the spindle speed to increase as the tool reaches the center of the part. The excessive speed can throw parts and damage tooling.*

## G97 Constant Surface Speed OFF (Group 13)

This commands the control to NOT adjust the spindle speed based on the diameter of cut and cancels any G96 command. When G97 is in effect, any **S** command is revolutions per minute (RPM).

## G98 Feed Per Minute (Group 10)

G98 changes how the **F** address code is interpreted. The value of **F** indicates inches per minute when Setting 9 is set to **INCH**, and **F** indicates millimeters per minute when Setting 9 is set to **MM**.

## G99 Feed Per Revolution (Group 10)

This command changes how the **F** address is interpreted. The value of **F** indicates inches per revolution of the spindle when Setting 9 is set to **INCH**, while **F** indicates millimeters per revolution of the spindle when Setting 9 is set to **MM**.

## G100 Disable / G101 Enable Mirror Image (Group 00)

\***X** - X-axis command

\***Z** - Z-axis command

\* indicates optional. At least one is required.

Programmable mirror image can be turned on or off individually for the X and/or Z Axis. The bottom of the screen indicates when an axis is mirrored. These G codes are used in a command block without any other G codes and do not cause any Axis motion. G101 turns on mirror image for any Axis listed in that block. G100 turns off mirror image for any Axis listed in the block. The actual value given for the X or Z code has no effect; G100 or G101 by itself has no effect. For example, G101 X 0 turns on X-axis mirror.



**NOTE:**

*Settings 45 and 47 may be used to manually select mirror image.*

## G103 Limit Block Look-Ahead (Group 00)

G103 specifies the maximum number of blocks the control looks ahead (Range 0-15), for example:

```
G103 [P..] ;
```

During machine motions, the control prepares future blocks (lines of code) ahead of time. This is commonly called “Block Look-ahead.” While the control executes the current block, it has already interpreted and prepared the next block for continuous motion.

A program command of G103 P0, or simply G103, disables block limiting. A program command of G103 Pn limits look-ahead to n blocks.

G103 is useful for debugging macro programs. The control interprets Macro expressions during look-ahead time. If you insert a G103 P1 into the program, the control interprets macro expressions (1) block ahead of the currently executing block.

It is best to add several empty lines after a G103 P1 is called. This ensures that no lines of code after the G103 P1 are interpreted until they are reached.

G103 affects cutter compensation and High Speed Machining.



**NOTE:**

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

## G105 Servo Bar Command

This is the G-code used to command a Bar Feeder.

G105 [In.nnnn] [Jn.nnnn] [Kn.nnnn] [Pnnnnn] [Rn.nnnn]

- I** - Optional Initial Push Length (macro variable #3101) Override (variable #3101 if I is not commanded)
- J** - Optional Part Length + Cutoff (macro variable #3100) Override (variable #3100 if J is not commanded)
- K** - Optional Min Clamping Length (macro variable #3102) Override (variable #3102 if K is not commanded)
- P** - Optional cutoff subprogram
- R** - Optional spindle orientation for new bar

I, J, K are overrides to macro variable values listed on the Current Commands Page. The control applies the override values only to the command line in which they are located. The values stored in Current Commands are not modified.

## G110 / G111 Coordinate System #7/#8 (Group 12)

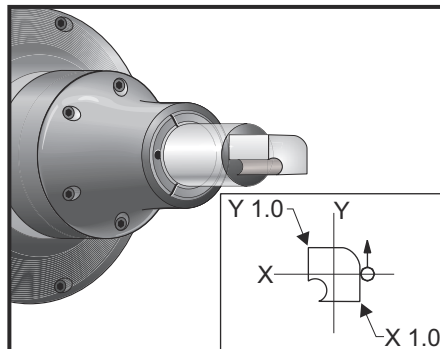
G110 selects #7 and G111 selects #8 additional work offset coordinates. All subsequent references to axes positions are interpreted in the new work offset coordinate system. Operation of G110 and G111 is the same as G154 P1 and G154 P2.

## G112 XY to XC Interpolation (Group 04)

The G112 XY to XC coordinate interpolation feature lets you program subsequent blocks in Cartesian XY coordinates, which the control automatically converts to polar XC coordinates. While it is active, the control uses the G17 XY for G01 linear XY strokes and G02 and G03 for circular motion. G112 also converts X, Y position commands into rotary C-Axis and linear X-axis moves.

### G112 Program Example

**F7.49:** G112 XY to XC Interpolation



```

%
o61121 (G112 XY TO XC INTERPOLATION) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an end mill) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G17 (Call XY plane) ;
G98 (Feed per min) ;
M154 (Engage C Axis) ;
P1500 M133 (Live tool CW at 1500 RPM) ;
G00 G54 X0.875 C0. Z0.1 (Rapid to 1st position) ;
G112 (XY to XC interpretation);
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G1 Z0. F15. (Feed towards face) ;
Y0.5 F5. (Linear feed) ;
G03 X.25 Y1.125 R0.625 (Feed CCW) ;
G01 X-0.75 (Linear feed) ;
G03 X-0.875 Y1. R0.125 (Feed CCW) ;
G01 Y-0.25 (Linear Feed) ;
G03 X-0.75 Y-0.375 R0.125 (Feed CCW) ;
G02 X-0.375 Y-0.75 R0.375 (Feed CW) ;
G01 Y-1. (Linear feed) ;
G03 X-0.25 Y-1.125 R0.125 (Feed CCW) ;
G01 X0.75 (Linear feed) ;
G03 X0.875 Y-1. R0.125 (Feed CCW) ;
G01 Y0. (Linear feed) ;
G00 Z0.1 (Rapid retract) ;
(BEGIN COMPLETION BLOCKS) ;
G113 (Cancel G112) ;
M155 (Disengage C Axis) ;
M135 (Live tool off) ;
G18 (Return to XZ plane) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 (Z home) ;
M30 (End program) ;

```

## **G113 Cancel XY to XC Interpolation (Group 04)**

G113 cancels the Cartesian to Polar coordinate conversion.

## G114-G129 Coordinate System #9-#24 (Group 12)

G114 - G129 codes are user-settable coordinate systems, #9 - #24, for work offsets. All subsequent references to axes' positions are interpreted in the new coordinate system. Work coordinate system offsets are entered from the **Active Work Offset** display page. Operation of G114 - G129 codes is the same as G154 P3 - G154 P18.

## G154 Select Work Coordinates P1-P99 (Group 12)

This feature provides 99 additional work offsets. G154 with a P value from 1 to 99 activates additional work offsets. For example G154 P10 selects work offset 10 from the list of additional work offsets.



### NOTE:

*G110 to G129 refer to the same work offsets as G154 P1 through P20; they can be selected by using either method.*

When a G154 work offset is active, the heading in the upper right work offset will show the G154 P value.



### NOTE:

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

### G154 work offsets format

#14001-#14006 G154 P1 (also #7001-#7006 and G110)  
#14021-#14026 G154 P2 (also #7021-#7026 and G111)  
#14041-#14046 G154 P3 (also #7041-#7046 and G112)  
#14061-#14066 G154 P4 (also #7061-#7066 and G113)  
#14081-#14086 G154 P5 (also #7081-#7086 and G114)  
#14101-#14106 G154 P6 (also #7101-#7106 and G115)  
#14121-#14126 G154 P7 (also #7121-#7126 and G116)  
#14141-#14146 G154 P8 (also #7141-#7146 and G117)  
#14161-#14166 G154 P9 (also #7161-#7166 and G118)  
#14181-#14186 G154 P10 (also #7181-#7186 and G119)  
#14201-#14206 G154 P11 (also #7201-#7206 and G120)

#14221-#14221 G154 P12 (also #7221-#7226 and G121)  
#14241-#14246 G154 P13 (also #7241-#7246 and G122)  
#14261-#14266 G154 P14 (also #7261-#7266 and G123)  
#14281-#14286 G154 P15 (also #7281-#7286 and G124)  
#14301-#14306 G154 P16 (also #7301-#7306 and G125)  
#14321-#14326 G154 P17 (also #7321-#7326 and G126)  
#14341-#14346 G154 P18 (also #7341-#7346 and G127)  
#14361-#14366 G154 P19 (also #7361-#7366 and G128)  
#14381-#14386 G154 P20 (also #7381-#7386 and G129)  
#14401-#14406 G154 P21  
#14421-#14426 G154 P22  
#14441-#14446 G154 P23  
#14461-#14466 G154 P24  
#14481-#14486 G154 P25  
#14501-#14506 G154 P26  
#14521-#14526 G154 P27  
#14541-#14546 G154 P28  
#14561-#14566 G154 P29  
#14581-#14586 G154 P30  
#14781-#14786 G154 P40  
#14981-#14986 G154 P50  
#15181-#15186 G154 P60  
#15381-#15386 G154 P70  
#15581-#15586 G154 P80  
#15781-#15786 G154 P90  
#15881-#15886 G154 P95  
#15901-#15906 G154 P96  
#15921-#15926 G154 P97



#15941-#15946 G154 P98

#15961-#15966 G154 P99

## G184 Reverse Tapping Canned Cycle For Left Hand Threads (Group 09)

**F** - Feed Rate in inches (mm) per minute

**R** - Position of the R plane

**S** - RPM, called prior to G184 is necessary

**\*W** - Z-axis incremental distance

**\*X** - X-axis motion command

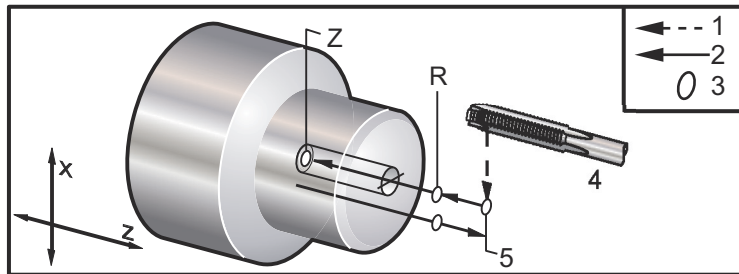
**Z** - Position of bottom of hole

\* indicates optional

Programming Notes: When tapping, the feedrate is the lead of the thread. See example of G84, when programmed in G99 Feed per Revolution.

It is not necessary to start the spindle CCW before this canned cycle; the control does this automatically.

**F7.50:** G184 Reverse Tapping Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Left-hand tap, [5] Starting plane, [R] R plane, [Z] Position of the bottom of the hole.



## G186 Reverse Live Tool Rigid Tap (For Left Hand Threads) (Group 09)

**F** - Feed Rate

**C** - C-Axis position

**R** - Position of the R plane

**S** - RPM, called prior to G186 is necessary

**W** - Z-axis incremental distance

**\*X** - Part Diameter X-axis motion command

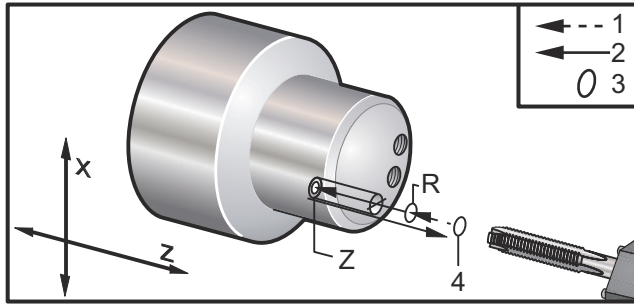
**\*Y** - Y-axis motion command

**Z** - Position of bottom of hole

\* indicates optional

**F7.51:**

G95, G186 Live Tooling Rigid Tapping: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting plane, [R] R plane, [Z] Position of the bottom of the hole.



It is not necessary to start the spindle CW before this canned cycle; the control does this automatically. See G84.

## G187 Accuracy Control (Group 00)

G187 is an accuracy command that can set and control both the smoothness and max corner rounding value when cutting a part. The format for using G187 is G187 Pn Ennnn.

**P** - Controls the smoothness level, P1(rough), P2(medium), or P3(finish). Temporarily overrides Setting 191.

**E** - Sets the max corner rounding value. Temporarily overrides Setting 85.

Setting 191 sets the default smoothness to the user specified **ROUGH**, **MEDIUM**, or **FINISH** when G187 is not active. The **Medium** setting is the factory default setting.


**NOTE:**

*Changing Setting 85 to a low value may make the machine operate as if it is in exact stop mode.*


**NOTE:**

*Changing setting 191 to **FINISH** will take longer to machine a part. Use this setting only when needed for the best finish.*

G187 Pm Ennnn sets both the smoothness and max corner rounding value. G187 Pm sets the smoothness but leaves max corner rounding value at its current value. G187 Ennnn sets the max corner rounding but leaves smoothness at its current value. G187 by itself cancels the E value and sets smoothness to the default smoothness specified by Setting 191. G187 will be canceled whenever **[RESET]** is pressed, M30 or M02 is executed, the end of program is reached, or **[EMERGENCY STOP]** is pressed.

## G195 / G196 Forward/Reverse Live Tool Radial Tapping (Diameter) (Group 09)

**F** - Feed Rate per revolution (G99)

**\*U** - X-Axis incremental distance to the bottom of hole

**S** - RPM, called prior to G195

**X** - X- Axis absolute position at the bottom of hole

**\*Z** - Z-Axis absolute position motion command

**R** - Position of the R plane

**\*C** - C-Axis absolute motion command

**\*Y** - Y-Axis absolute motion command

**\*W** - Z-Axis incremental motion command

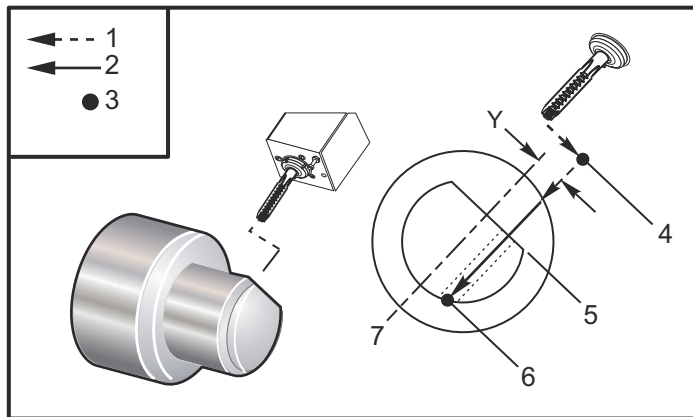
**\*E** - Chip-clean RPM (Spindle reverses to remove chips after each hole)

\* indicates optional

This G-code is modal in that it activates the canned cycle until it is canceled or another canned cycle is selected. The cycle begins from the current position, tapping to the X-Axis depth specified. An R plane can be used.

S RPM should be called out as a positive number. It is not necessary to start the spindle in the correct direction; the control does this automatically.

**F7.52:** G195/G196 Live Tooling Rigid Tapping: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting point, [5] Part surface, [6] Bottom of the hole, [7] Centerline.



```
o61951 (G195 LIVE RADIAL TAPPING) ;  
(G54 X0 Y0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is a tap) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
M154 (Engage C Axis) ;
```

```

G00 G54 X3.25 Z-0.75 C0. (Start Point) ;
M08 (coolant on) ;
(BEGIN CUTTING BLOCK) ;
S500 (Select tap RPM) ;
G195 X2. F0.05 (Taps to X2., bottom of hole) ;
G00 C180. (Index C-Axis) ;
G00 C270. Y-1. Z-1. (Index C-Axis, YZ-axis positioning) ;
G80 (Cancel Canned Cycle);
(BEGIN COMPLETION BLOCKS) ;
G00 Z0.25 M09 (Rapid retract, coolant off) ;
M155 (Disengage C Axis) ;
G53 X0 Y0 (X & Y home) ;
G53 Z0 (Z home) ;
M30 (End program) ;

```

## **G198 Disengage Synchronous Spindle Control (Group 00)**

G198 disengages synchronous spindle control and allows independent control of the main spindle and the secondary spindle.

## **G199 Engage Synchronous Spindle Control (Group 00)**

**\*R** - Degrees, phase relationship of following spindle to commanded spindle

\* indicates optional

This G code synchronizes the RPM of the two spindles. Position or speed commands to the following spindle, usually the secondary spindle, are ignored when spindles are in synchronous control. However, M codes on the two spindles are controlled independently.

The spindles remain synchronized until synchronous mode is disengaged using G198. This is the case even if power is cycled.

An R value on the G199 block positions the following spindle to a specified number of degrees, relative to the 0 mark on the commanded spindle. Examples of R values in G199 blocks:

```

G199 R0.0 (The following spindle's origin, 0-mark, matches the
commanded spindle's origin, 0-mark) ;
G199 R30.0 (The following spindle's origin, 0-mark, is
positioned +30 degrees from the commanded spindle's origin,
0-mark) ;
G199 R-30.0 (The following spindle's origin, 0-mark, is
positioned -30 degrees from the commanded spindle's origin,
0-mark) ;

```

When an R value is specified on the G199 block, the control first matches the velocity on the following spindle to that of the commanded spindle, then adjusts the orientation (R value in the G199 block). Once the specified R orientation is achieved the spindles are locked in synchronous mode until disengaged with a G198 command. This can also be achieved at zero RPM. Refer also to the G199 portion of the Synchronized Spindle Control Display on 188.

```
%
o61991 (G199 SYNC SPINDLES) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;

(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G20 G40 G80 G99 (Safe startup) ;

G00 G54 X2.1 Z0.5 ;
G98 M08 (Feed per min, turn coolant on) ;

(BEGIN CUTTING BLOCKS) ;
G01 Z-2.935 F60. (Linear feed) ;
M12 (Air blast on) ;
M110 (Secondary spindle chuck clamp) ;
M143 P500 (Secondary spindle to 500 RPM) ;
G97 M04 S500 (Main spindle to 500 RPM) ;
G99 (Feed per rev) ;
M111 (Secondary spindle chuck unclamp) ;
M13 (Air blast off) ;
M05 (main spindle off) ;
M145 (Secondary spindle off) ;
G199 (Synch spindles) ;

G00 B-28. (Rapid secondary spindle to face of part) ;
G04 P0.5 (Dwell for .5 sec) ;
G00 B-29.25 (Feed secondary spindle onto part) ;
M110 (secondary spindle chuck clamp) ;
G04 P0.3 (Dwell for .3 sec) ;
M08 (Turn coolant on) ;
G97 S500 M03 (Turn spindle on at 500 RPM, CSS off) ;
G96 S400 (CSS on, RPM is 400) ;
G01 X1.35 F0.0045 (Linear feed) ;
X-.05 (Linear feed) ;
G00 X2.1 M09 (Rapid retract) ;
G00 B-28. (Rapid secondary spindle to face of part) ;
G198 (Synch spindle off) ;
```

```

M05 (Turn off main spindle) ;
G00 G53 B-13.0 (Secondary spindle to cut position);
G00 G53 X-1. Y0 Z-11. (Rapid to 1st position) ;
(*****second side of part*****)
G55 G99 (G55 for secondary spindle work offset) ;
G00 G53 B-13.0 ;
G53 G00 X-1. Y0 Z-11. ;
G14 ;
T101 (Select tool and offset 1) ;
G50 S2000 (limit spindle to 1000 RPM);
G97 S1300 M03 ( ;
G00 X2.1 Z0.5 ;
Z0.1 M08 ;
G96 S900 ;
G01 Z0 F0.01 ;
X-0.06 F0.005 ;
G00 X1.8 Z0.03 ;
G01 Z0.005 F0.01 ;
X1.8587 Z0 F0.005 ;
G03 X1.93 Z-0.0356 K-0.0356 ;
G01 X1.935 Z-0.35 ;
G00 X2.1 Z0.5 M09 ;
G97 S500 ;
G15 ;
G53 G00 X-1. Y0 Z-11. ;

(BEGIN COMPLETION BLOCKS) ;
G00 G53 X0 M09 (X home) ;
G53 Z0 (Z home) ;
G28 H0. (Unwind C-Axis) ;
M30 (End program) ;
%
```

## G200 Index on the Fly (Group 00)

**U** - Optional relative move in X to tool change position

**W** - Optional relative move in Z to tool change position

**X** - Optional final X position

**Z** - Optional final Z position

**T** - Required tool number and offset number in standard form

G200 Index on the Fly causes the lathe to perform a move away, change tools, and move back to the part, to save time.

**CAUTION:**

*The G200 does speed things up, but it also requires you to be more careful. Make sure you proof the program well, at 5% rapid, and be very cautious if you are starting from the middle of the program.*

Normally, your tool change line consists of a few lines of code, like:

```
G53 G00 X0. (BRING TURRET TO SAFE X TC POS) ;  
G53 G00 Z-10. (BRING TURRET TO SAFE Z TC POS) ;  
T202 ;
```

Using G200, changes this code to:

```
G200 T202 U.5 W.5 X8. Z2. ;
```

If T101 just finished turning the O.D. of the part, you don't need to go back to a safe tool change position, when using a G200. Instead (as in the example) the moment the G200 line is called the turret:

1. Unclamps, in its current position.
2. Moves incrementally in the X and Z axes by the values stated in U and W (U.5 W.5)
3. Completes the tool change at this position.
4. Using the new tool and work offsets, it rapids to the XZ position called out on the G200 line (X8. Z2.).

This all happens very quickly, and nearly all at the same time, so try it out a few times, away from the chuck.

When the turret unclamps, it moves towards the spindle a tiny amount (perhaps .1-.2"), so you do not want the tool directly up against your jaws or collet when the G200 is commanded.

Because the U and W moves are incremental distances from where the tool is currently, if you hand jog away and start your program in a new position, the turret moves up and to the right of that new position. In other words, if you manually jogged back within .5" of your tailstock, and then commanded G200 T202 U.5 W1. X1. Z1., the turret would hit your tailstock - moving an incremental W1. (1" to right). For this reason, you may want to setup your Setting 93 and Setting 94, Tailstock Restricted Zone.

Information on this can be found on page 111.

## G211 Manual Tool Setting / G212 Auto Tool Setting

**T** - Tool number. May be entered as `Tnn` or `Tnnnn`.

**H** - Tool tip direction. `H-5` will approach the probe from the X (-) side and `H5` from the X (+) side.

**\*K** - Indicates a calibration cycle. (Values 1 or 2)

**\*M** - Tool breakage tolerance value.

**\*C** - Drill diameter value. Only valid with tip directions 5-8. Offset will be adjusted by half this amount (i.e. the program assumes a 90-degree drill point).

**\*X** - Adjust the approach and start points of a probing cycle.

**\*Z** - Adjust the approach and start points of a probing cycle.

**\*B** - Allows the user to use a different amount to move the tool the tool in X or Z while probing (from the start point to in position over the probe). Default value is 6mm.

**\*U** - Adjust the X start point on `H1` - 4.

**\*W** - Adjust the Z start point on `H1` - 4.

\*indicates optional



### NOTE:

*The G211 code requires a Tnnn code, either directly before the G211line, or on the same line. The G211 code also requires an Hnnn code. The G212 code only requires an Hnnn code on the same line but a Tnnn code tool call is required prior.*

### Using G211 Manual Tool Setting

**IMPORTANT:** *The Automatic Tool Probe must be calibrated before using G211 / G212.*

The `G211` code is used to set an initial tool offset (X, Z or both). To use the probe arm must be lowered. Then the tool tip jogged into place about 0.25 in from the corner of the problem that corresponds to the desired tip direction. The code will either use the current tool offset if one has been called previously or the tool offset may be chosen using a `T` code. The cycle will probe the tool, enter the offset and return the tool to the start position.

### Using G212 Auto Tool Setting

The `G212` code is used to re-probe a tool that already has an offset set, such after an insert is changed. It can be also be used to check for tool breakage. The tool will be moved from any location into proper orientation to the probe by the `G212` command. This path is determined by the tool tip direction variable `H`, this variable must be correct or the tool may crash.

**IMPORTANT:** *Care must be used for touching off any back working tools, to keep from hitting the spindle or the back wall of the machine. A tool and offset must be called Tnnn before running G212, or an alarm will be generated.*



G212 code is used to re-probe a tool that already has an offset set, such after an insert is changed. It can be also be used to check for tool breakage. The tool will be moved from any location into proper orientation to the probe by the G212 command. This path is determined by the tool tip direction variable H and it must be correct or the tool may crash.

**IMPORTANT:** Care must be used for touching off any back working tools, to keep from hitting the spindle or the back wall of the machine. A tool and offset must be called Tnnn before running G212, or an alarm will be generated.

## G241 Radial Drill Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**R** - Position of the R plane (diameter)

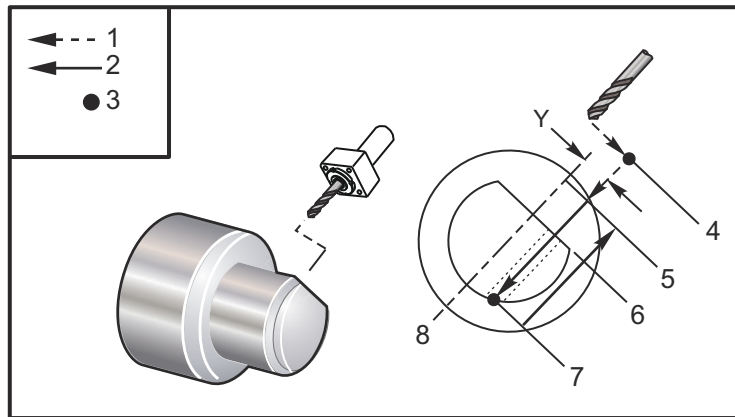
**X** - Position of bottom of hole (diameter)

**\*Y** - Y-axis absolute motion command

**\*Z** - Z-axis absolute motion command

\* indicates optional

**F7.53:** G241 Radial Drill Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting point, [5] R plane, [6] Part surface, [Z] Bottom of the hole, [8] Centerline.



```
%
o62411 (G241 RADIAL DRILLING) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a drill) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per min) ;
M154 (Engage C Axis) ;
```

```

G00 G54 X5. Z-0.75 (Rapid to 1st position) ;
P1500 M133 (Live tool CW at 1500 RPM) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G241 X2.1 Y0.125 Z-1.3 C35. R4. F20. (Begin G241) ;
X1.85 Y-0.255 Z-0.865 C-75. (next position) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z1. M09 (Rapid retract, coolant off) ;
M155 (Disengage C Axis) ;
M135 (Live tool off) ;
G53 X0 Y0 (X & Y Home) ;
G53 Z0 (Z Home) ;
M30 (End program) ;
%
```

## G242 Radial Spot Drill Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**P** - The dwell time at the bottom of the hole

**R** - Position of the R plane (Diameter)

**X** - Position of bottom of hole (Diameter)

**\*Y** - Y-axis motion command

**\*Z** - Z-axis motion command

\* indicates optional

This G code is modal. It remains active until it is canceled (G80) or another canned cycle is selected. Once activated, every motion of Y and/or Z executes this canned cycle.

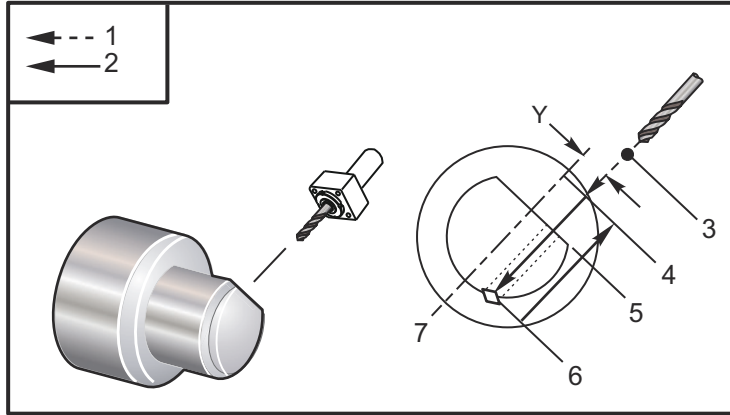


**NOTE:**

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

**F7.54:**

G242 Radial Spot Drill Canned Cycle: [1] Rapid, [2] Feed, [3] Starting point, [4] R plane, [5] Part surface, [6] Dwell at the bottom of the hole, [7] Centerline.



%

```

o62421 (G242 RADIAL SPOT DRILL) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a spot drill) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per min) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P1500 M133 (Live tool CW at 1500 RPM) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G241 X2.1 Y0.125 Z-1.3 C35. R4. P0.5 F20. ;
(Drill to X2.1) ;
X1.85 Y-0.255 Z-0.865 C-75. P0.7 (next position) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z1. M09 (Rapid retract, coolant off) ;
M155 (Disengage C Axis) ;
M135 (Live tool off) ;
G53 X0 Y0 (X & Y Home) ;
G53 Z0 (Z Home) ;
M30 (End program) ;

```

%

## G243 Radial Normal Peck Drilling Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**\*I** - Size of first cutting depth

**\*J** - Amount to reduce cutting depth each pass

**\*K** - Minimum depth of cut

**\*P** - The dwell time at the bottom of the hole

**\*Q** - The cut-in value, always incremental

**R** - Position of the R plane (Diameter)

**X** - Position of bottom of hole (Diameter)

**\*Y** - Y-axis absolute motion command

**\*Z** - Z-axis absolute motion command

\* indicates optional

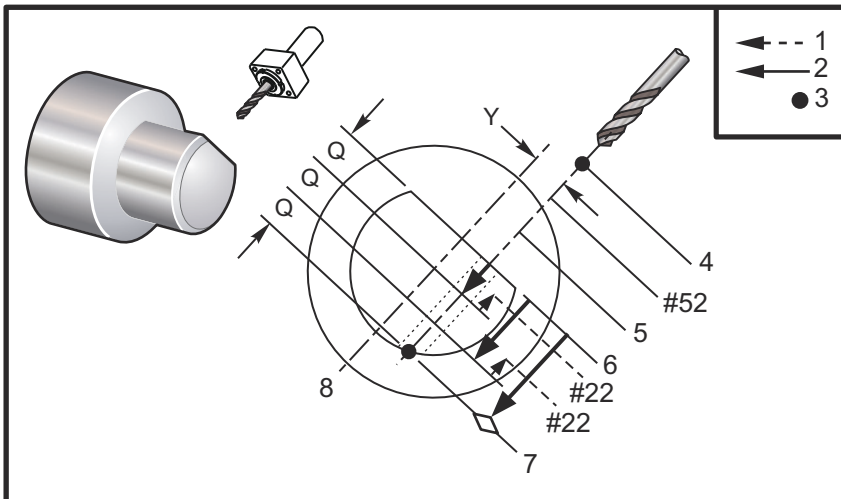


**NOTE:**

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

**F7.55:**

G243 Radial Normal Peck Drilling Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] R plane, [#52] Setting 52, [5] R plane, [6] Part surface, [#22] Setting 22, [7] Dwell at the bottom of the hole, [8] Centerline.



Programming Notes: If **I**, **J**, and **K** are specified, a different operating mode is selected. The first pass will cut in the value of **I**, each succeeding cut will be reduced by amount **J**, and the minimum cutting depth is **K**. Do not use a **Q** value when programming with **I**, **J**, and **K**.

Setting 52 changes the way G243 works when it returns to the R plane. Usually the R plane is set well outside the cut to insure that the chip clearing motion allows the chips to clear the hole. However, this is wasted motion when first drilling through this empty space. If Setting 52 is set to the distance required to clear chips, the R plane can be put much closer to the part being drilled. When the clear move to R occurs, the Z will be moved past R by this value in setting 52. Setting 22 is the amount to feed in X to get back the same point at which the retraction occurred.

```
%
o62431 (G243 RADIAL PECK DRILL CYCLE) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a drill) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per min) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P1500 M133 (Live tool CW at 1500 RPM) ;
M08 (Coolant on) ;
G243 X2.1 Y0.125 Z-1.3 C35. R4. Q0.25 F20. ;
(Drill to X2.1) ;
X1.85 Y-0.255 Z-0.865 C-75. Q0.25 (Next position);
G00 Z1. (Rapid retract) ;
M135 (Live tool off) ;
G00 G53 X0 M09(X home, coolant off) ;
G53 Z0 ;
M00 ;
(G243 - RADIAL WITH I,J,K PECK DRILLING) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P1500 M133 (Live tool CW - 1500 RPM) ;
M08 (Coolant on) ;
G243 X2.1 Y0.125 Z-1.3 I0.25 J0.05 K0.1 C35. R4. F5. ;
(Drill to X2.1) ;
X1.85 Y-0.255 Z-0.865 I0.25 J0.05 K0.1 C-75. ;
(next position) ;
(BEGIN COMPLETION BLOCKS) ;
M155 (Disengage C Axis) ;
M135 (Turn live tool off) ;
G00 G53 X0 Y0 M09 (X & Y home, coolant off) ;
G53 Z0 (Z home) ;
M30 (End program) ;
%
```

## G245 Radial Boring Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**R** - Position of the **R** plane (Diameter)

**X** - Position of bottom of hole (Diameter)

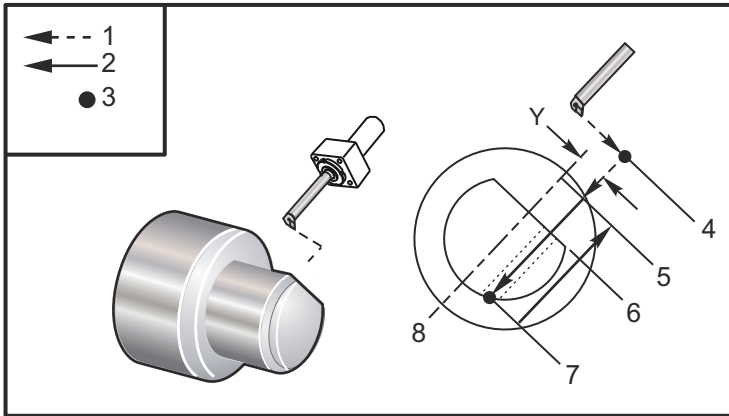
**\*Y** - Y-axis absolute motion command

**\*Z** - Z-axis absolute motion command

\* indicates optional

**F7.56:**

G245 Radial Boring Canned Cycle: [1] Rapid, [2] Feed, [3] Start or end of stroke, [4] Starting point, [5] R plane, [6] Part surface, [7] Bottom of the hole, [8] Centerline.



%

```

o62451 (G245 RADIAL BORING) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a boring tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per min) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P500 M133 (Live tool CW at 500 RPM) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G245 X2.1 Y0.125 Z-1.3 C35. R4. F20. ;
(Bore to X2.1) ;
X1.85 Y-0.255 Z-0.865 C-75. (next position) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z1. M09 (Rapid retract, coolant off) ;
M155 (Disengage C Axis) ;
    
```

```

M135 (live tool off) ;
G53 X0 Y0 (X & Y home) ;
G53 Z0 (Z home) ;
M30 (End program) ;
%
```

## **G246 Radial Bore and Stop Canned Cycle (Group 09)**

**C** - C-Axis absolute motion command

**F** - Feed Rate

**R** - Position of the R plane (Diameter)

**X** - Position of bottom of hole (Diameter)

**\*Y** - Y-axis absolute motion command

**\*Z** - Z-axis absolute motion command

\*indicates optional

This G code stops the spindle once the tool reaches the bottom of the hole. The tool is retracted once the spindle has stopped.

```

%
o62461 (G246 RADIAL BORE AND STOP) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a boring tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per min) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P500 M133 (Live tool CW at 500 RPM) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G246 X2.1 Y0.125 Z-1.3 C35. R4. F20. ;
(Bore to X2.1) ;
X1.85 Y-0.255 Z-0.865 C-75. (next position) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z1. M09 (Rapid retract, coolant off) ;
M155 (Disengage C Axis) ;
M135 (Live tool off) ;
G53 X0 Y0 (X & Y Home) ;
G53 Z0 (Z Home) ;
M30 (End program) ;
%
```

## G247 Radial Bore and Manual Retract Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**R** - Position of the R plane (Diameter)

**\*X** - Position of bottom of hole (Diameter)

**\*Y** - Y-axis absolute motion command

**\*Z** - Z-axis absolute motion command

\* indicates optional

This G code stops the spindle at the bottom of the hole. At this point the tool is manually jogged out of the hole. The program continues when **[CYCLE START]** is pressed.

```
%
O62471 (G247 RADIAL BORE AND MANUAL RETRACT) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a boring tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per minute) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P500 M133 (Live tool CW at 500 RPM) ;
M08 (coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G247 X2.1 Y0.125 Z-1.3 C35. R4. F20. ;
(Bore to X2.1) ;
X1.85 Y-0.255 Z-0.865 C-75. (next position) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z1. M09 (Rapid retract, Coolant off) ;
M155 (Disengage C Axis) ;
M135 (Live tool off) ;
G53 X0 Y0 (X & Y Home) ;
G53 Z0 (Z Home) ;
M30 (End program) ;
%
```



## G248 Radial Bore and Dwell and Manual Retract Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**P** - The dwell time at the bottom of the hole

**R** - Position of the R plane (Diameter)

**\*X** - Position of bottom of hole (Diameter)

**\*Y** - Y-axis absolute motion command

**\*Z** - Z-axis absolute motion command

\* indicates optional

This G code stops the tool at the bottom of the hole, and dwells with the tool turning for the time designated with the **P** value. At this point the tool is manually jogged out of the hole. The program continues when **[CYCLE START]** is pressed.

```
%
o62481 (G248 RADIAL BORE, DWELL, MANUAL RETRACT) ;
(G54 X0 Y0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is a boring tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G98 (Feed per minute) ;
M154 (Engage C Axis) ;
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;
P500 M133 (Live tool CW at 500 RPM) ;
M08 (coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G248 X2.1 Y0.125 Z-1.3 C35. R4. P1. F20. ;
(Bore to X2.1) ;
X1.85 Y-0.255 Z-0.865 C-75. (next position) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z1. M09 (Rapid retract, coolant off) ;
M155 (Disengage C Axis) ;
M135 (Live tool off) ;
G53 X0 Y0 (X & Y Home) ;
G53 Z0 (Z Home) ;
M30 (End program) ;
%
```

## G249 Radial Bore and Dwell Canned Cycle (Group 09)

**C** - C-Axis absolute motion command

**F** - Feed Rate

**P** - The dwell time at the bottom of the hole

**R** - Position of the R plane

**X** - Position of bottom of hole

**\*Y** - Y-axis motion command

**\*Z** - Z-axis motion command

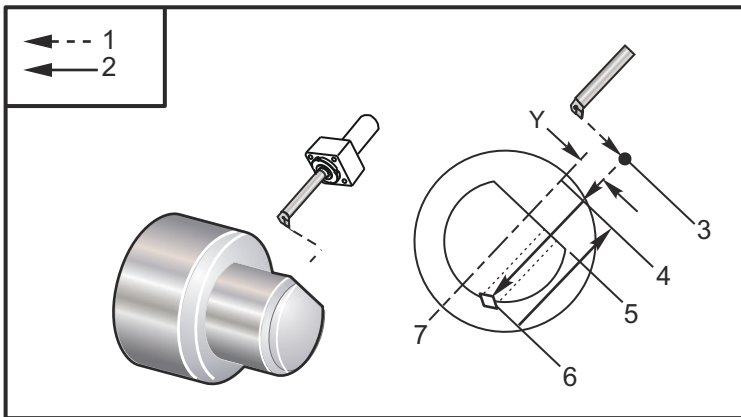
\* indicates optional

  
**NOTE:**

*The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

**F7.57:**

G249 Radial Bore and Dwell Canned Cycle: [1] Rapid, [2] Feed, [3] Starting point, [4] R plane, [5] Part surface, [6] Dwell at the bottom of the hole, [7] Centerline.



␣

```
o62491 (G249 RADIAL BORE AND DWELL) ;  
(G54 X0 Y0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is a boring tool) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
G98 (Feed per minute) ;  
M154 (Engage C Axis) ;  
G00 G54 X5. Y0.125 Z-1.3 (Rapid to 1st position) ;  
P500 M133 (Live tool CW at 500 RPM) ;  
M08 (coolant on) ;
```